

Manas Tripathi

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EDUCATION

Indian Institute of Information Technology, Ranchi
Bachelor of Technology in Computer Science and Engineering

- CGPA: 9.30/10.0

Aug 2023 to present

EXPERIENCE

Summer Research Intern
Indian Institute of Technology, Gandhinagar

May 2025 to Jul 2025

- Researched real-time Bias Correction of Ensemble Rainfall Forecasts (ERFS) under **Prof. Vimal Mishra** using **Knowledge Distillation** to improve flood prediction.
- Engineered a **Knowledge Distillation** framework training **XGBoost** on **LSTM-derived features**, without historical observations across water-stressed agricultural regions.
- Delivered significant performance gains: **+0.57** in **NSE**, **32.7%** reduction in **MAE**, and **18%** lower **PBIAS** compared to baseline, quantifiably strengthening flood forecast reliability for climate-vulnerable river basins.
- Drives accurate rainfall forecasting, advancing early warning systems and climate-resilient operational decisions.

PROJECTS

MindScope | Python, NLP, BERT, TensorFlow

[GitHub](#)

- Fine-tuned BERT on Reddit's mental health dataset to detect stress and anxiety from user narratives - targeting 1 in 5 adults globally who lack access to timely mental health screening.
- Achieved **65% accuracy** at **~100ms inference latency**, enabling scalable, low-cost mental health screening for underserved populations where clinician access averages **1 per 10,000 people**.

EmotionGAN | Python, Deep Learning, VGG16, cGAN, TensorFlow

[GitHub](#)

- Pre-trained **VGG16** on facial expression data, achieving **58% emotion detection accuracy** across 7 distinct emotional states for real-world affective computing applications.
- Analyzed artwork emotions via the trained model and built a **Conditional GAN** generating matched emotional images, fusing affective computing with generative synthesis for therapeutic art applications.

VoiceEmote | Python, Deep Learning, Librosa, Speech Processing

[GitHub](#)

- Built a **Librosa**-powered speech model extracting **MFCCs**, **chroma**, and **spectral features** to classify emotions from raw audio, applicable to mental health monitoring and assistive voice technology.
- Reached **72% training accuracy**, unlocking deployable emotion-aware voice assistants serving **466M+ people** living with hearing or speech-related mental health challenges globally.

VisionGAN | Python, Deep Learning, DCGAN, CV

[GitHub](#)

- Developed a **DCGAN** generating clinically realistic glaucoma images, expanding scarce ophthalmic datasets and sharpening diagnostic models for **80M+ glaucoma patients** at risk of preventable blindness.

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS

ML & DL Domains: Computer Vision, NLP, Audio Processing, GANs, RAG

Libraries & Frameworks: TensorFlow, PyTorch, Keras, scikit-learn, LangChain, Librosa, Pandas, NumPy, Matplotlib

CS Fundamentals: Data Structures, Algorithms, System Design, Software Engineering, Computer Networks, Operating Systems

Developer Tools: Git, GitHub, Jupyter, Google Colab, Linux Terminal

ACHIEVEMENTS

- Secured **All India Rank 6499** in **GATE CS 2026** - top **2.5% nationwide** among all appearing candidates.
- Advanced through **Smart India Hackathon (SIH) 2024** institute selection, engineering an **AI-powered job seeker-recruiter platform** solving graduate unemployment under **Government of Rajasthan's** Smart Education initiative.